Comparison, Comparison, Comparison

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“The fundamental analytical act in statistical reasoning is to answer the question ´compared to what?´

Whether we are evaluating changes over space or time, searching big data bases, adjusting and controlling for variables, designing experiments, specifying multiple regressions, or doing just about any kind of evidence-based reasoning, the essential point is to make intelligent and appropriate comparisons.

Thus visual displays, if they are to assist thinking, should show comparisons.”
(Tufte 2006: 127)

Some Visual Comparisons

Comparing before and after

Comparing to a standard

Comparing to context

Comparing by subgroups

Comparing to an implicit model (“what do we expect to see?”)

Some Graphical Principles for Comparisons

Should the comparison be made within a single graph or across multiple graphs?

If multiple graphs, make sure to use:

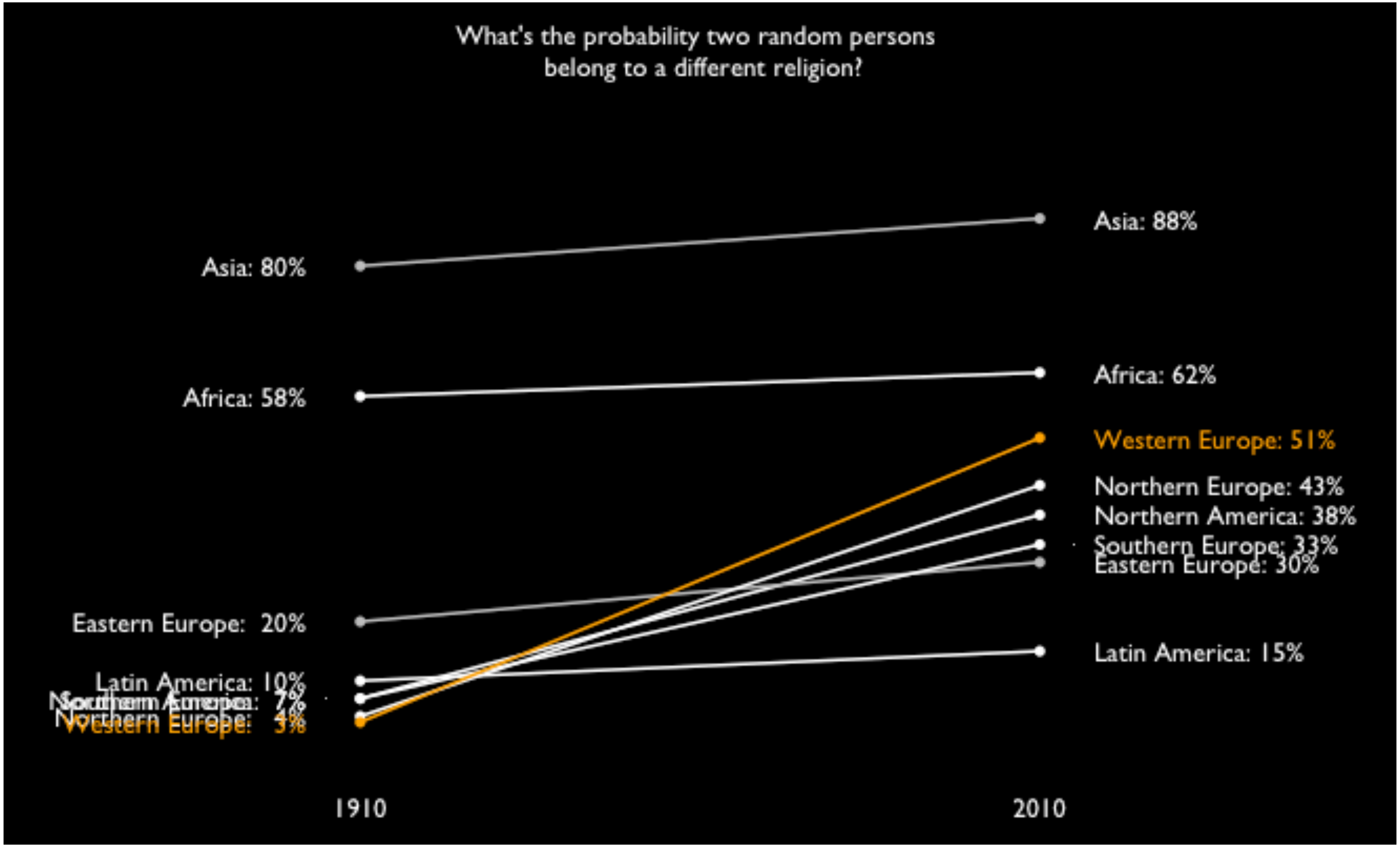
Common graph size

Common scales

Helpful alignment

Colour and shape can be used to compare groups within and across graphs.

Comparing before and after: Slope Graph

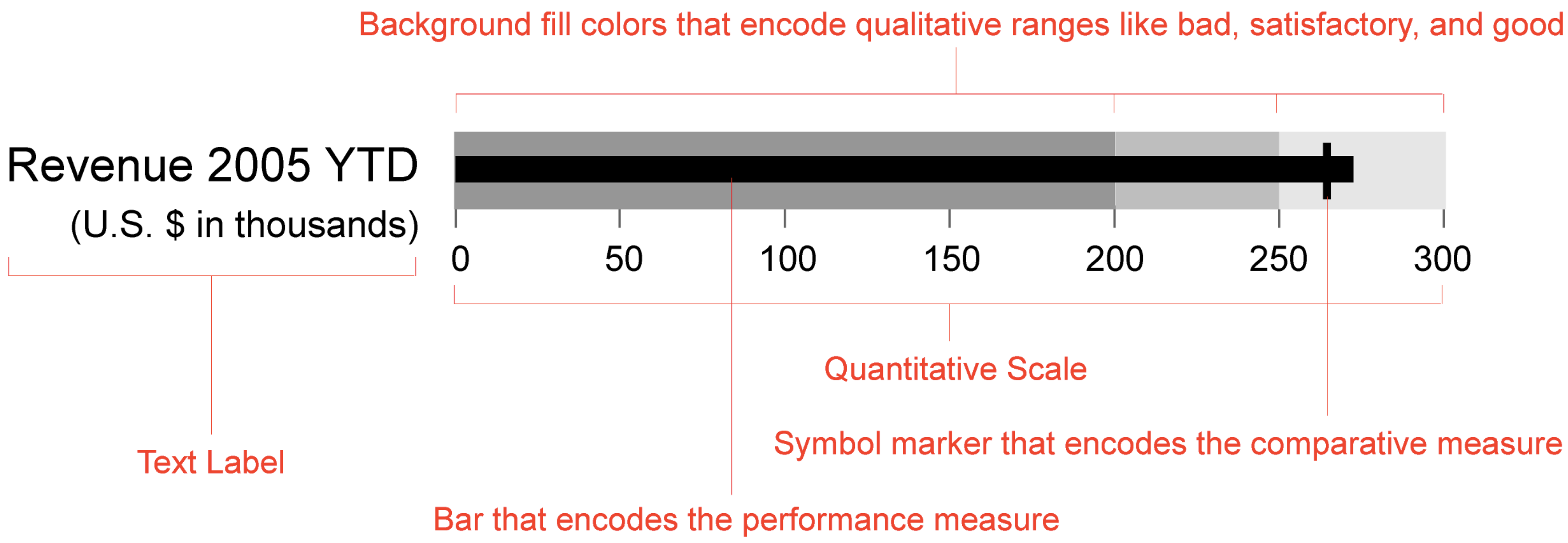


Comparing to a Standard: Bullet Graph

Bullet graphs compare a quantitative measure to

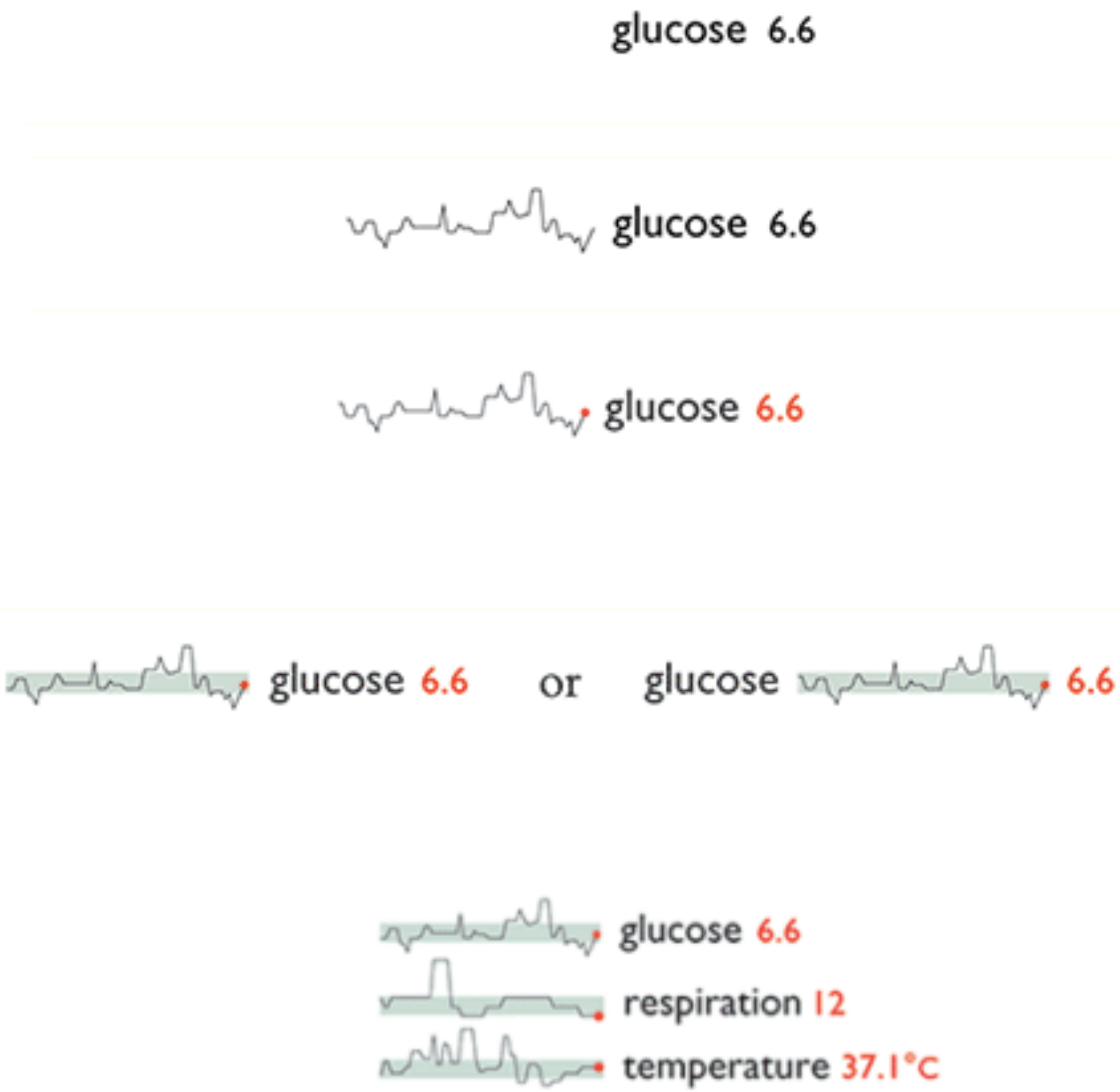
a) one or more related measures (e.g., a target or the same measure in the past) and

b) relate the measure to defined ranges that declare its qualitative state (e.g. good, satisfactory, and poor).



Comparing to a Standard: Bullet Graph

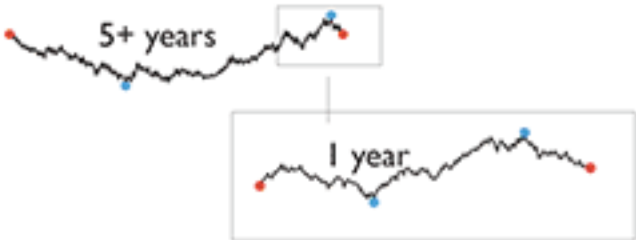
Comparing before and after/to context/to a standard: Spark Lines



Comparing before and after/to context/to a standard: Spark Lines

		2003.4.28	12 months	2004.4.28	low	high
Euro foreign exchange	$	1.1025		1.1907	1.0783	1.2858

		1999.1.1	65 months	2004.4.28	low	high		2003.4.28	12 months	2004.4.28	low	high
Euro foreign exchange	$	1.1608		1.1907	.8252	1.2858	$	1.1025		1.1907	1.0783	1.2858
Euro foreign exchange	¥	121.32		130.17	89.30	140.31	¥	132.54		130.17	124.80	140.31
Euro foreign exchange	£	0.7111		0.6665	.5711	0.7235	£	0.6914		0.6665	0.6556	0.7235



Comparing by Subgroups: Small Multiples

"High-density graphics help us to compare parts of the data by displaying much information within the view of the eye: we look at one page at a time and the more on the page, the more effective and comparative our eye can be." (Tufte 2001: 168).

Data Density = number of entries in data matrix/area of data graphic

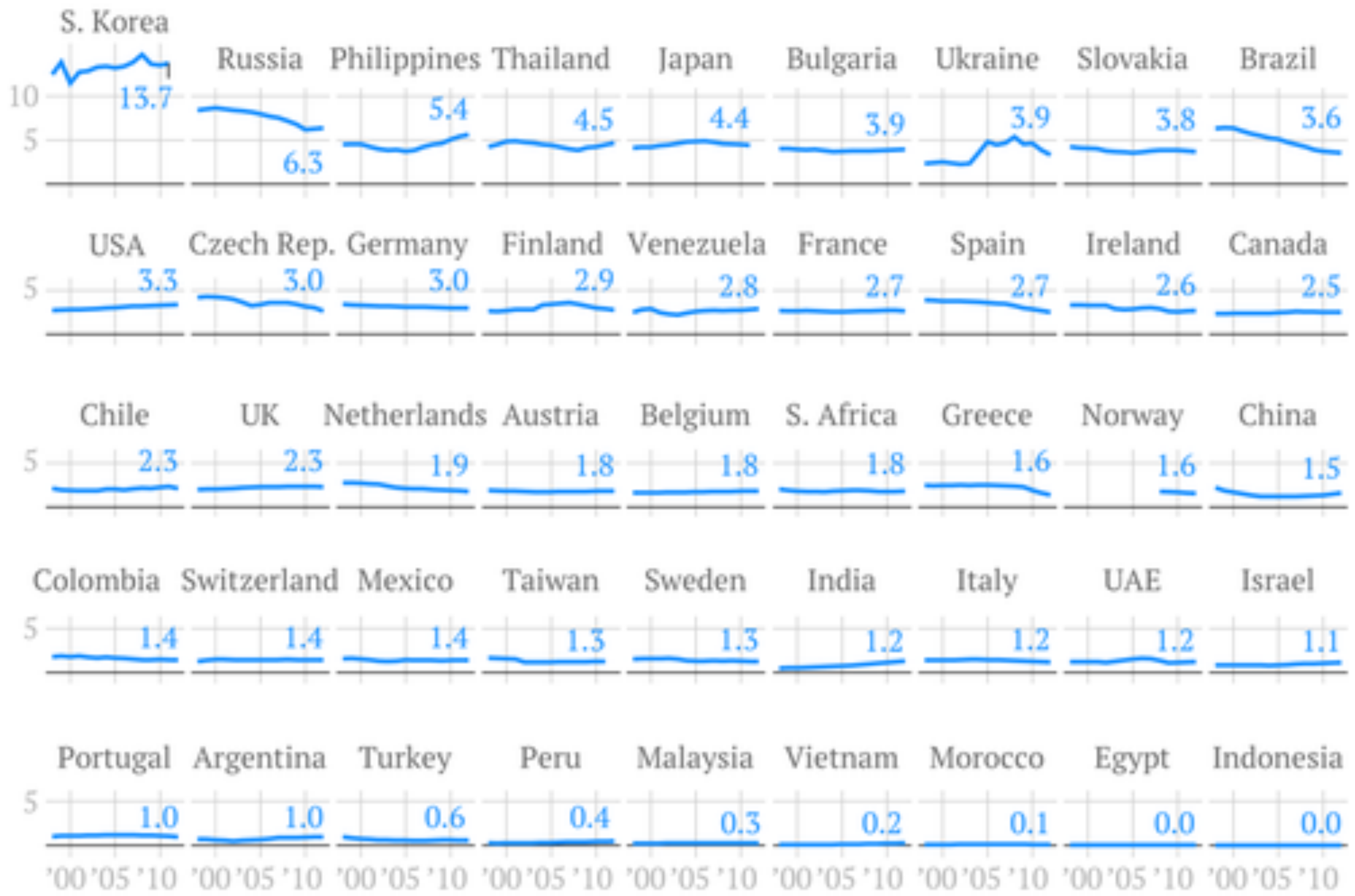
Maximize data density and the size of the data matrix, within reason.

Graphics can be shrunk way down.

This yields one of the most powerful visualization formats: The Small Multiple.

The average amount of liquor consumed by a person of drinking age

Shots per week of any spirit

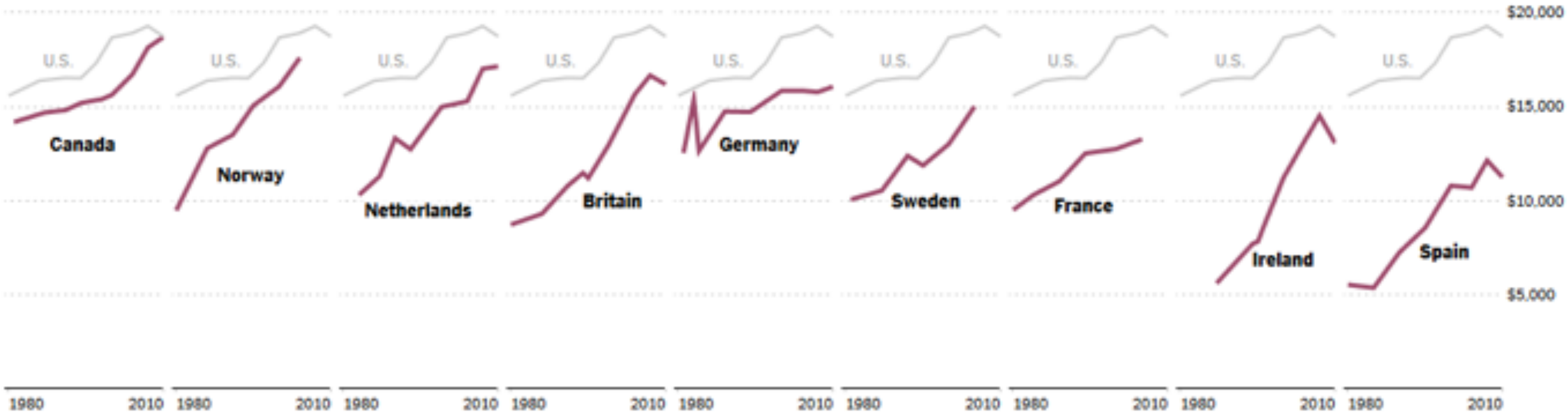


Quartz | Ritchie King

Data: Euromonitor

The United States' once-strong lead in middle class incomes is shrinking.

MEDIAN PER CAPITA INCOME AFTER TAXES



Source: New York Times/Luxembourg Income Study analysis

2000: State-level support (orange) or opposition (green) on school vouchers, relative to the national average of 45% support

Orange and green colors correspond to states where support for vouchers was greater or less than the national average.
The seven ethnicreligious cagetories are mutually exclusive. "Evangelicals" includes Mormons as well as born-again Protestants.
Where a category represents less than 1% of the voters of a state, the state is left blank.